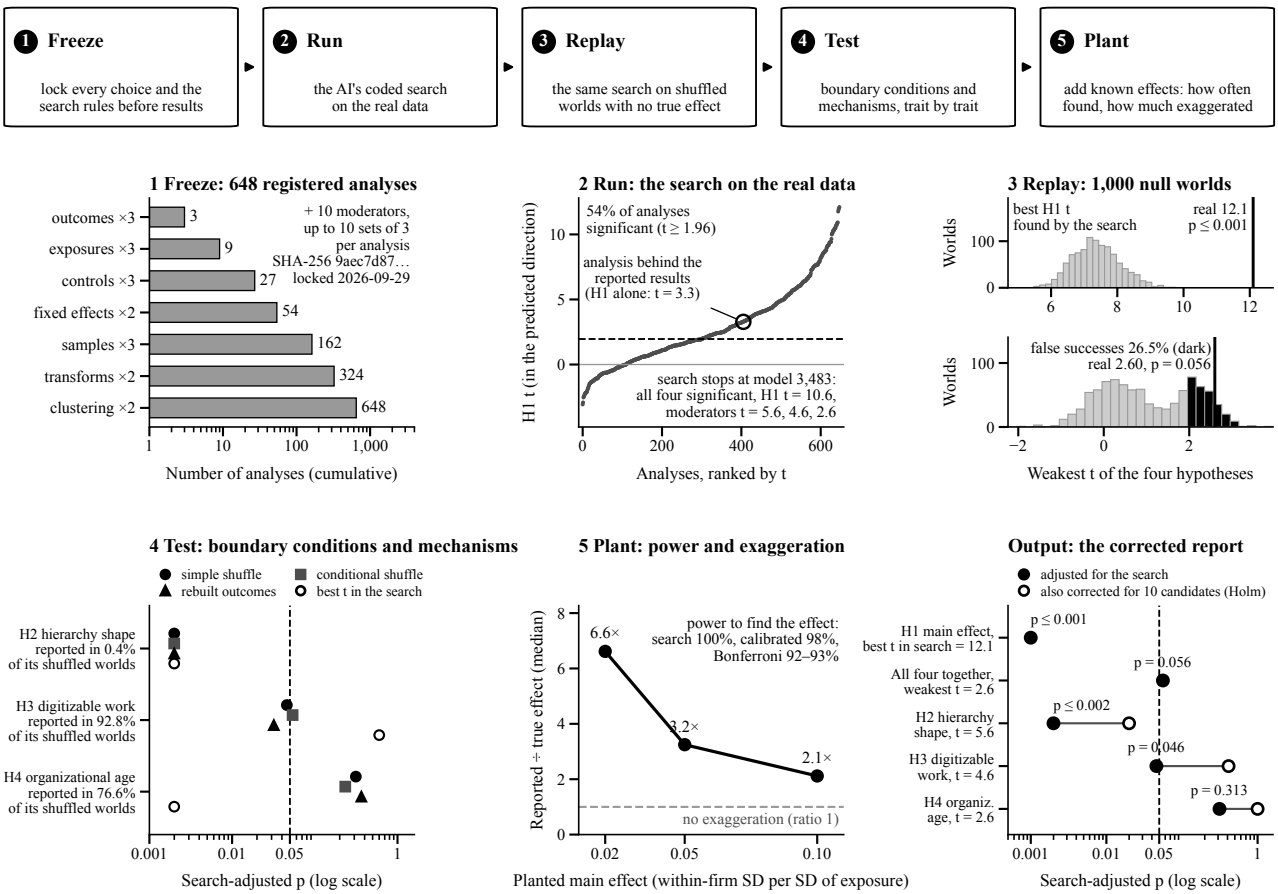

1 Freeze
lock every choice and the search rules before results
2 Run
the AI's coded search on the real data
3 Replay
the same search on shuffled worlds with no true effect
4 Test
boundary conditions and mechanisms, trait by trait
5 Plant
add known effects: how often found, how much exaggerated
1 Freeze: 648 registered analyses
outcomes ×3
exposures ×3
controls ×3
fixed effects ×2
samples ×3
transforms ×2
clustering ×2
3
9
27
54
162
324
648
+ 10 moderators, up to 10 sets of 3 per analysis
SHA-256 9aec7d87…
locked 2026-09-29
1
10
100
1,000
Number of analyses (cumulative)
2 Run: the search on the real data
54% of analyses significant (t ≥ 1.96)
analysis behind the reported results (H1 alone: t = 3.3)
search stops at model 3,483: all four significant, H1 t = 10.6, moderators t = 5.6, 4.6, 2.6
H1 t (in the predicted direction)
10
5
0
0
200
400
600
Analyses, ranked by t
3 Replay: 1,000 null worlds
best H1 t found by the search
real 12.1
p ≤ 0.001
Worlds
100
0
6
8
10
12
false successes 26.5% (dark)
real 2.60, p = 0.056
Worlds
100
0
−2
0
2
Weakest t of the four hypotheses
4 Test: boundary conditions and mechanisms
simple shuffle
conditional shuffle
rebuilt outcomes
best t in the search
H2 hierarchy shape reported in 0.4% of its shuffled worlds
H3 digitizable work reported in 92.8% of its shuffled worlds
H4 organizational age reported in 76.6% of its shuffled worlds
0.001
0.01
0.05
1
Search-adjusted p (log scale)
5 Plant: power and exaggeration
power to find the effect: search 100%, calibrated 98%, Bonferroni 92–93%
6.6×
3.2×
2.1×
Reported ÷ true effect (median)
8
6
4
2
0
no exaggeration (ratio 1)
0.02
0.05
0.10
Planted main effect (within-firm SD per SD of exposure)
Output: the corrected report
adjusted for the search
also corrected for 10 candidates (Holm)
H1 main effect, best t in search = 12.1
p ≤ 0.001
All four together, weakest t = 2.6
p = 0.056
H2 hierarchy shape, t = 5.6
p ≤ 0.002
H3 digitizable work, t = 4.6
p = 0.046
H4 organiz. age, t = 2.6
p = 0.313
0.001
0.01
0.05
1
Search-adjusted p (log scale)